GMI German-Malaysian Institute

DIPLOMA PROGRAMME

MINI PROJECT

Academic Period : July 2025

Code	*CIT 0633*	Course Name	*OPERATING SYSTEMS ADMINISTRATION*
Title	*Mini Project*	Examiner	*MOHD HAFIZ RAZALI*
Name (Individual)	ABDUL HARIZ BIN MOHD SARNUDIN (CBS24070208)	Program / Group	*CBS 1*

No	Assessment Criteria	CLO	PO	Level of diff	Full Marks	Score	Marks	Comment
1	Accuracy (Installation Windows Server operating system, client operating system, joining domain between server and client)	2	2	P3	20			
2	Functionality(Group Policies)	2	2	P3	10			
3	Functionality (Disk Quota)	2	2	P3	10			
4	Functionality (Home Directory)	2	2	P3	10			
5	Functionality (Configuring DHCP)	2	2	P3	20			
6	Functionality (Configuring Web server using www.YOURNAME.net)	2	2	P3	10			
7	Functionality (Configuring FTP server)	2	2	P3	10			
8	Documentation	3	9	A3	10			
				Total	100			
				Total Marks				
				PO 2				
				PO 4				
				Signature				

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NO.	CRITERIA	APPROACHES EXPECTATIONS M[1]	MEETS EXPECTATIONS M[2]	EXCELLENT M[3]	WEIGHT	MARKS
ASSIGNMENT (1) – 10%						
1	**Demonstration of tools**	Able to slightly demo the demonstration details	Able to demo most of the demonstration's details	Able to demo all of the demonstration's details	M/3*30	
2	**Interpreting Findings** (Analyses/concept knowledge)	Able to slightly interpret the findings and results	Able to interpret most of the findings and results	Able to interpret all of the findings and results	M/3*30	
3	**Fluency and Clarity** Presentation	None of the group members fluent and voice is not clear.	Most of the group members are fluent and voices are slightly easy to be heard clearly.	All group members are very fluent, and voices are easy to be heard clearly.	M/3*15	
4	**Defence ability/ presentation**	Not able to answer questions, not prepared and no confidence at all	Able to answer questions but with little preparation and confidence	Able to all answer questions very well and confidently. Very well prepared	M/3*15	
5	**Report Content & Format**	Report is poorly presented (unclear and confusing)	The report is clearly presented, but not enough.	The report is clearly presented and in details	M/3*10	
	TOTAL					

CIT 0633 OPERATING SYSTEMS ADMINISTRATION | Revision Number: Rev 12/2025 | Semester: July 2025

Vetting Form - Final Assessment (Mini Project)

Final Assignment

Raw Marks:	100
Percentage (in Doc 01):	30

No	Assessment Criteria	Full Marks	Score (if applicable)	Student's Marks	CLO	Level of Difficulty	DK. DP, NA (ETAC Programme Only)	Comments	PLO GMI: PLO CBS 9 PLO - Start Intake July 2023	PLO GMI: PLO MBOT 9 PLO	Vetters' Remark
1	Demonstration of Tools	30			2	P3			2	2	
2	Interpreting Findings (Analyses/concept knowledge)	30			2	P3			2	2	
3	Fluency and Clarity	15			3	A3			9	4	
4	Defence ability/ presentation	15			3	A3			9	4	
5	Report Content & Format	10			3	A3			9	4	
	Total Full Marks	**100**	**Total Student's Marks**			**0**					

SUMMARY

Note:
FM - Full Marks,
StM - Student's Mark

	CLO	Raw Marks FM	Raw Marks StM	Percentage FM	Percentage StM	PLO GMI: PLO CBS 9 PLO - Start Intake July 2023	Raw Marks FM	Raw Marks StM	Percentage FM	Percentage StM	PLO GMI: PLO MBOT 9 PLO	Raw Marks FM	Raw Marks StM	Percentage FM	Percentage StM
Total	CLO 1	0		0.0		PLO 1	0		0.0		PLO 3	0		0.0	
	CLO 2	60		18.0		PLO 2	60		18.0		PLO 2	60		18.0	
	CLO 3	40		12.0		PLO 9	40		12.0		PLO 4	40		12.0	
	CLO 4	0		0.0		#N/A	0		0.0		#N/A	0		0.0	
	CLO 5	0		0.0		#N/A	0		0.0		#N/A	0		0.0	
	CLO 6	0		0.0		#N/A	0		0.0		#N/A	0		0.0	
	Total Marks	**100**		**30.0**											

Summary of Vetter's Remark	Comply? YES	NO
Assessment		
Overall Distribution of Marks	YES	
Distribution of Marks per Item	YES	
Overall Assessment Items		
Format	YES	
CLO & PLO mapping per Item	YES	
Level of Difficulty	YES	
Clarity of Item	YES	
Correction Have Been Done	YES	

Vetter's Signature ( )

Vetter's Overall Remarks:

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SCENARIO

You have been assigned by your superior to work on a new project in developing a network server that applies Windows Server 2022 as an operating system. For the client they have to install Windows 10 as their new operating system.

During the installation, you have to make 2 partitions with the logical drive C: and drive D:

In this project, you have to configure as follows:
1. Create a new domain name system for the network that can resolve host names and IP addresses.

2. You have to create 3 OUs and each of the OU must have 5 users. Each OU should be personalized with specific wallpaper and logon hours for specific working days. You also have to assign 1 OU administrator to administer each OU.

3. Your superior also requested for each of the users to have their own disk quota to be set as 5MB per user. You have to show your quota works. Every user should be allocated with his/her own specific drive that is located in the server with full permission.

4. You have to assign for each of the users a specific home directory. In this case, when they log on to the network, they will have their own drive mapped to their home directory on the server.

5. In order to reduce the complexity of administering IP addresses for clients, you have to install and configure the DHCP server.

6. This company has its own website, and your superior intended to manage the website in the company's network server you are working on. In this case, you have to install and configure the web server. The website should be accessible by the user using the company domain name proceeding with www.

7. To enable transferring files between a client and server, you also have to install and configure the FTP server in the network server.

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Software used :

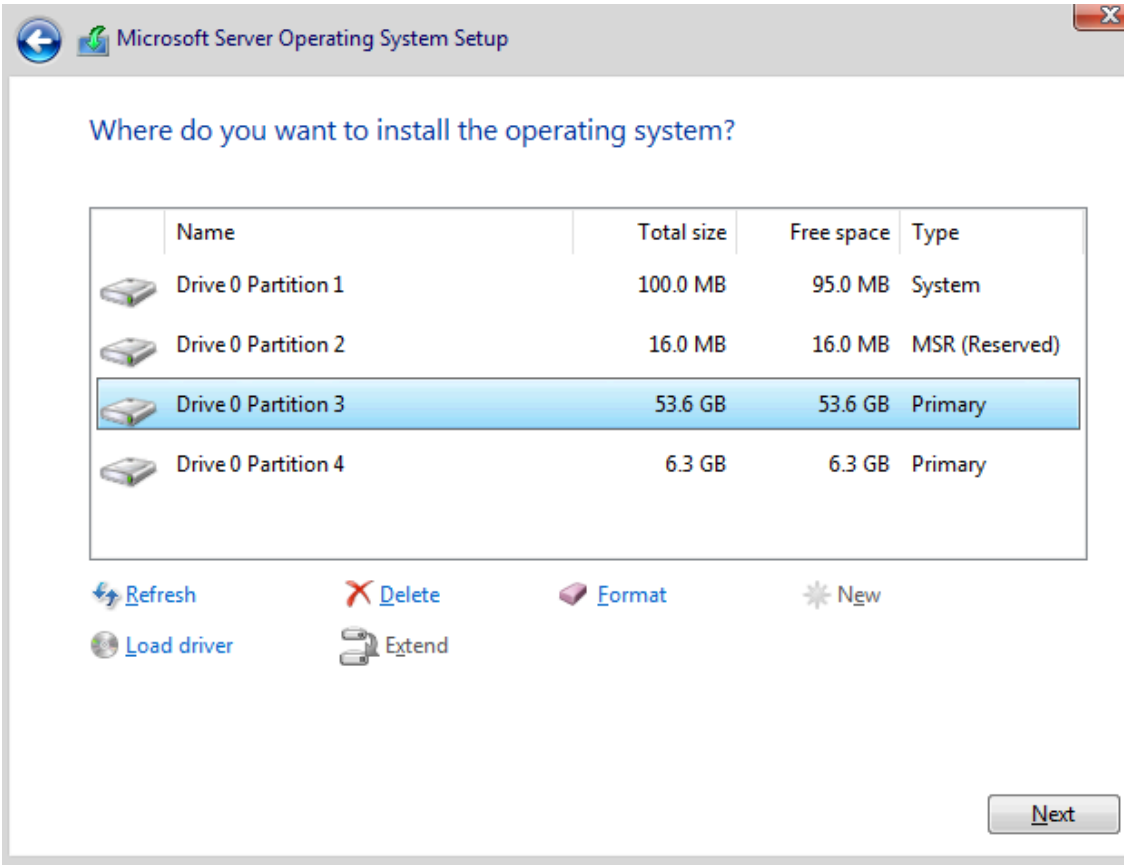
1) **VMware Workstation Pro 2025**
2) **Windows Server 2022 ISO File -> Server VM**
3) **Windows 10 ISO File -> Client VM**
4) **Gemini Ai (For Assistance)**

Hardware used :

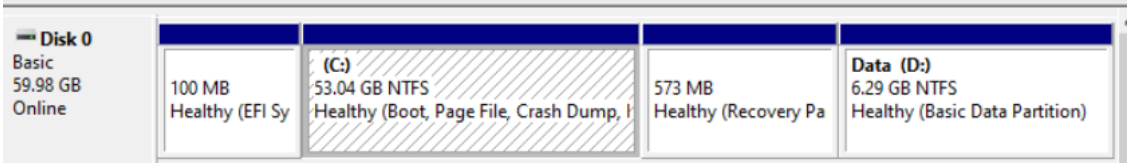
1) **Laptop -> Host Machine**
2) **Portable SSD (Stores VM Files)**

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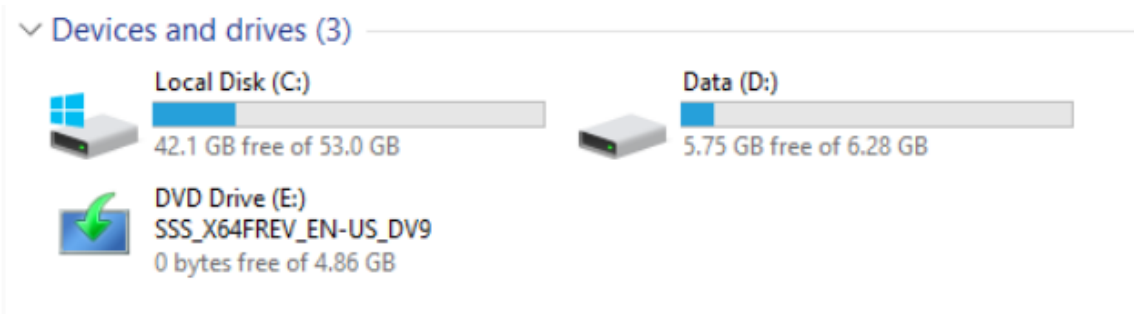
Disk Partitioning



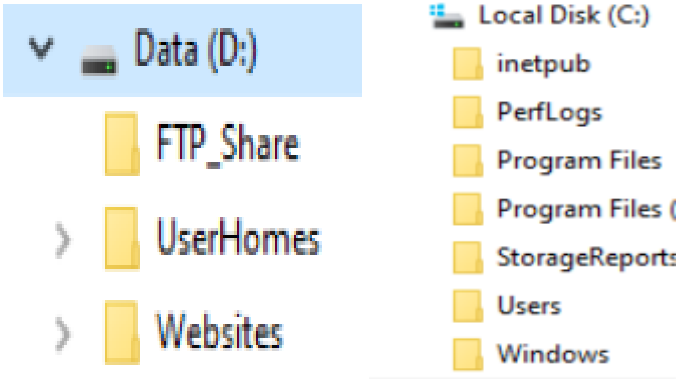
(D:) Drive can be set at Disk Management. Upon installation, the (D:) Drive was an unallocated disk.



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The server uses two disks: Local Disk (C:) for system data and configuration, and Data (D:) for user home folders, websites, and FTP directories

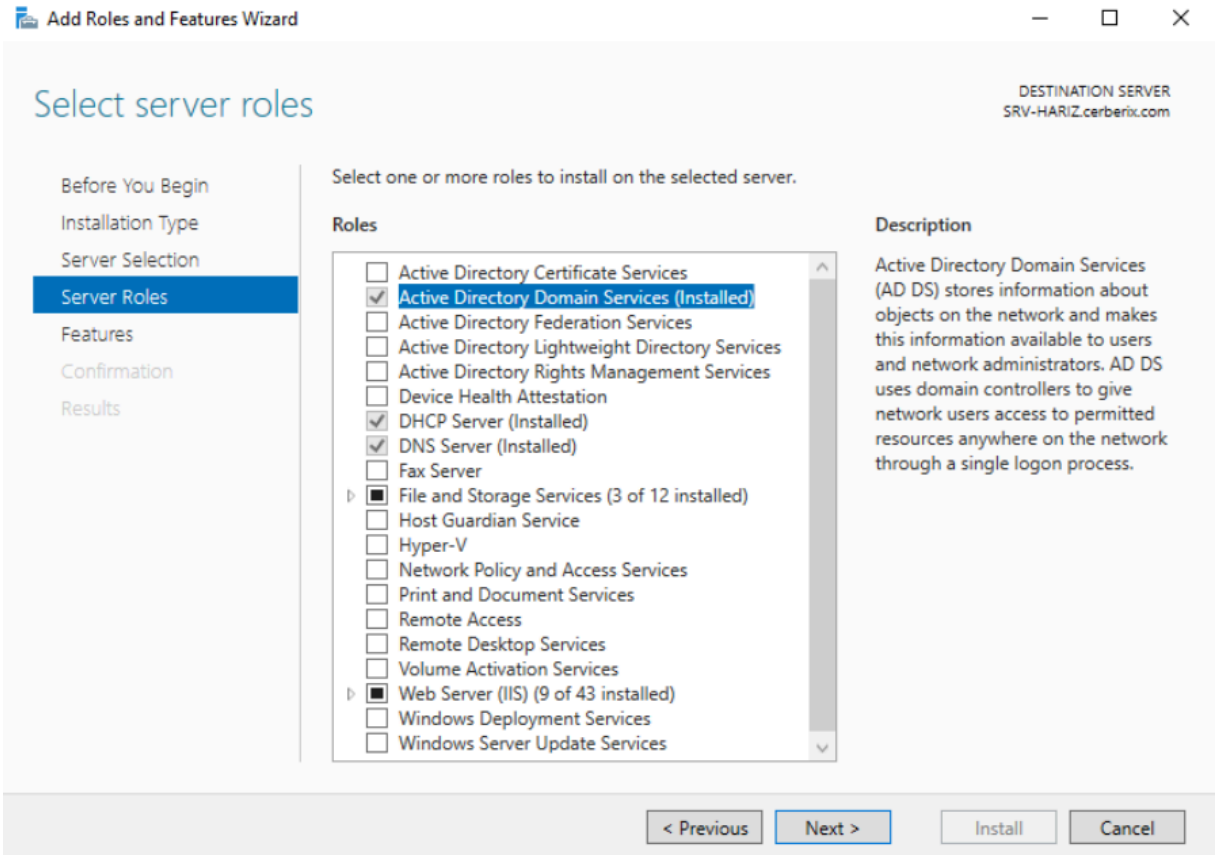


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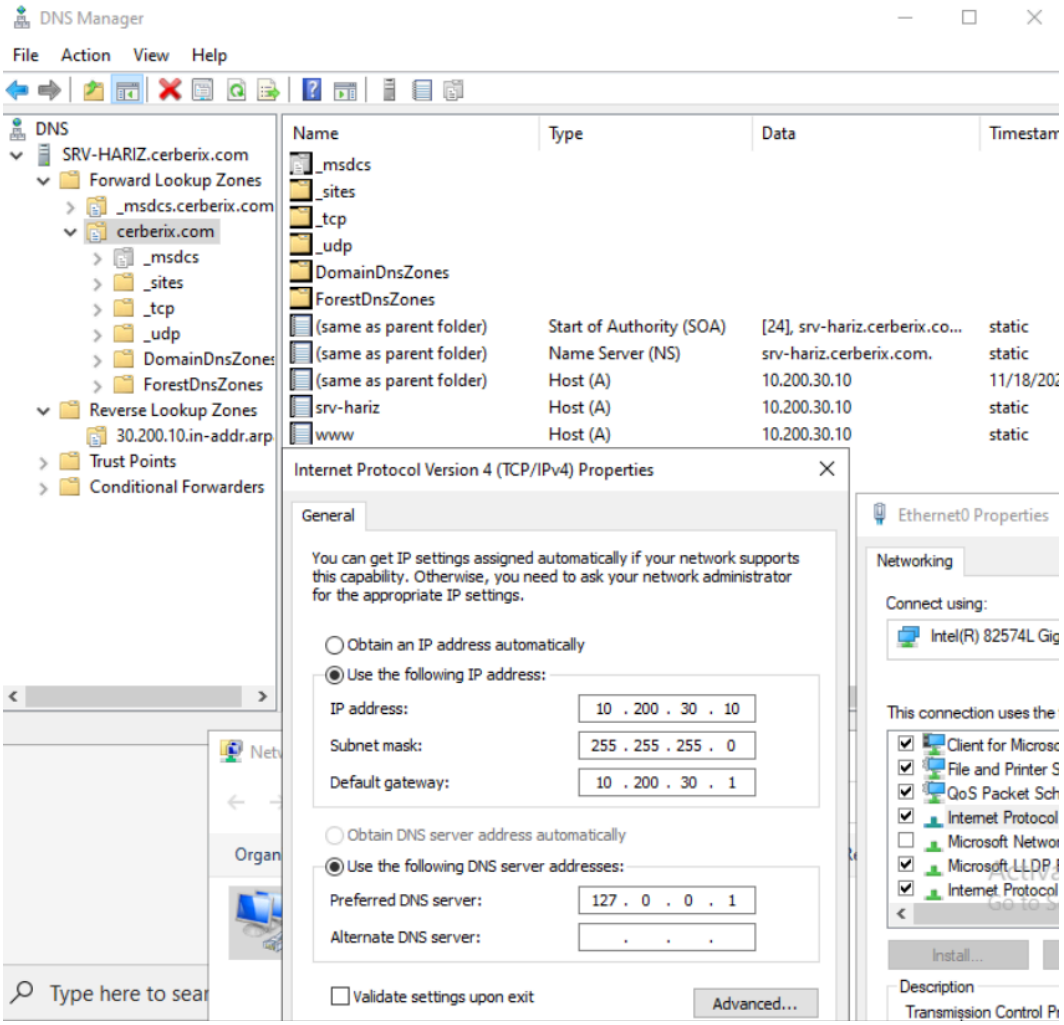
Domain Name System (DNS) & Active Directory Domain Services (AD-DS)

Domain : **www.cerberix.com**
Server IP : **10.200.30.10**
Default Gateway (Router) : **10.200.30.1**

DNS was configured automatically by installing and configuring AD-DS.



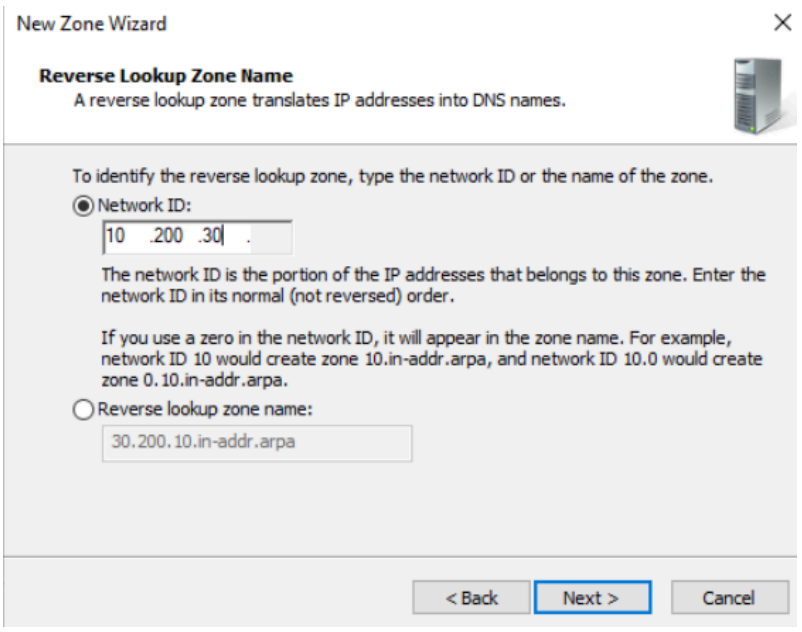
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Reverse Lookup must also be configured manually.





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Upon client joining the network through DHCP, a forward lookup zones will be automatically created in the DNS by AD.

Name	Type	Data	Timestamp
_msdcs			
_sites			
_tcp			
_udp			
DomainDnsZones			
ForestDnsZones			
(same as parent folder)	Start of Authority (SOA)	[30], srv-hariz.cerberix.co...	static
(same as parent folder)	Name Server (NS)	srv-hariz.cerberix.com.	static
(same as parent folder)	Host (A)	10.200.30.10	11/18/2025 4:00:00 PM
DESKTOP-ASL065R	Host (A)	10.200.30.5	11/18/2025 10:00:00 PM
DESKTOP-FVM9HLJ	Host (A)	10.200.30.6	11/19/2025 2:00:00 AM
srv-hariz	Host (A)	10.200.30.10	static
www	Host (A)	10.200.30.10	static

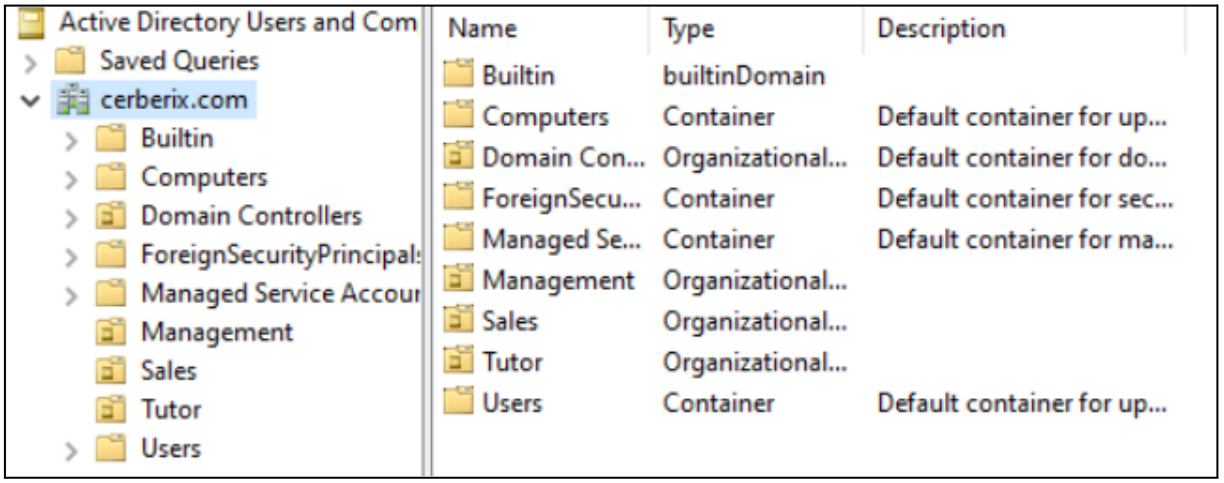
Same goes for the Reverse Lookup Zone.

Name	Type	Data	Timestamp
(same as parent folder)	Start of Authority (SOA)	[3], srv-hariz.cerberix.com...	static
(same as parent folder)	Name Server (NS)	srv-hariz.cerberix.com.	static
10.200.30.10	Pointer (PTR)	SRV-HARIZ.cerberix.com.	11/18/2025 6:00:00 PM
10.200.30.6	Pointer (PTR)	DESKTOP-FVM9HLJ.cerber...	11/19/2025 2:00:00 PM

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Organizational Unit

Overview



Created 3 Organizational Unit (OU), Management, Sales and Tutor. Management for the company (Cerberix) management team. Sales for the sales team and tutor for the teacher and tutor.

Cerberix is an educational company.

Management

Name	Type
Admin-Man...	User
Mng-1	User
Mng-2	User
Mng-3	User
Mng-4	User
Mng-5	User

Sales

Name	Type
Admin-Sales	User
User-1	User
User-2	User
User-3	User
User-4	User
User-5	User

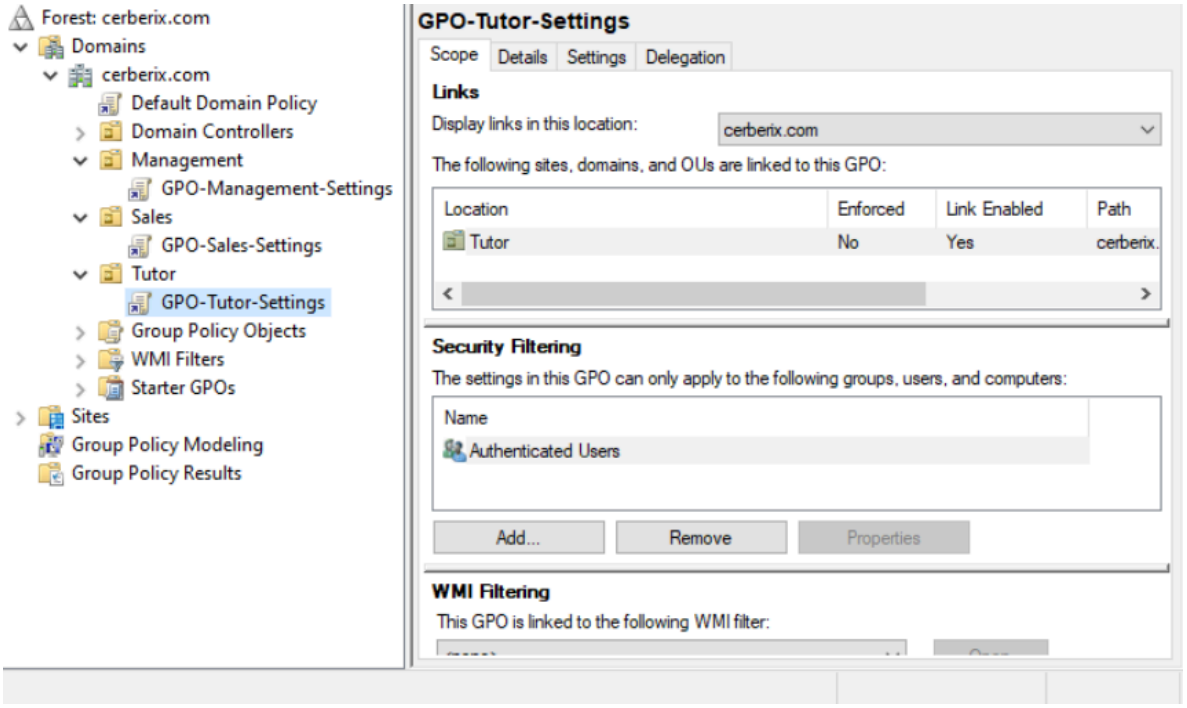
Tutor

Name	Type
Admin-Tutor	User
Tutor-1	User
Tutor-2	User
Tutor-3	User
Tutor-4	User
Tutor-5	User

Each administrator of an Organizational Unit (OU) is granted full delegate control over that OU.

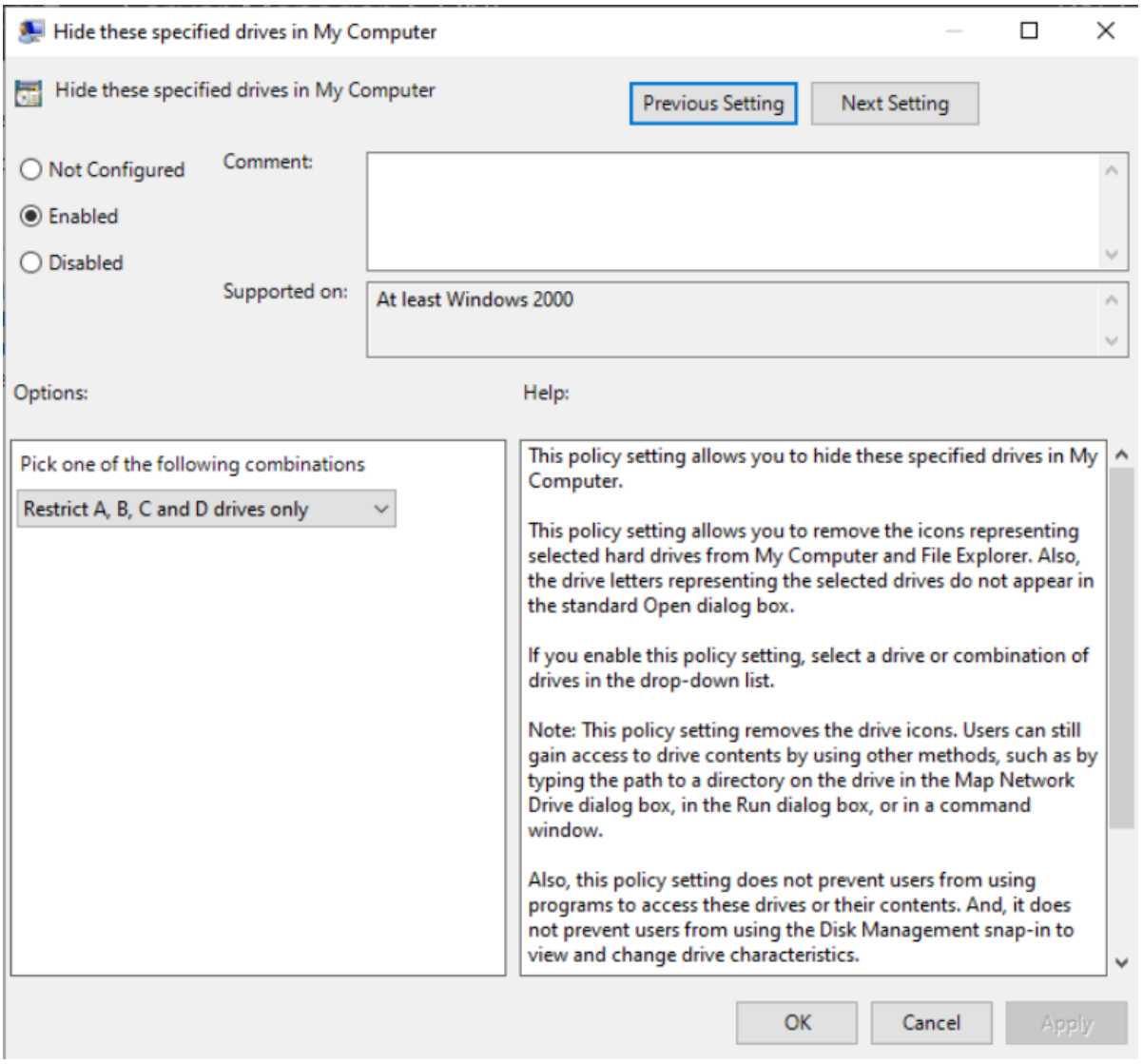
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Each Department also has their own Group Policy Management. Which the function is for assigning default wallpapers to each department. Basically set universal rules for the whole department.



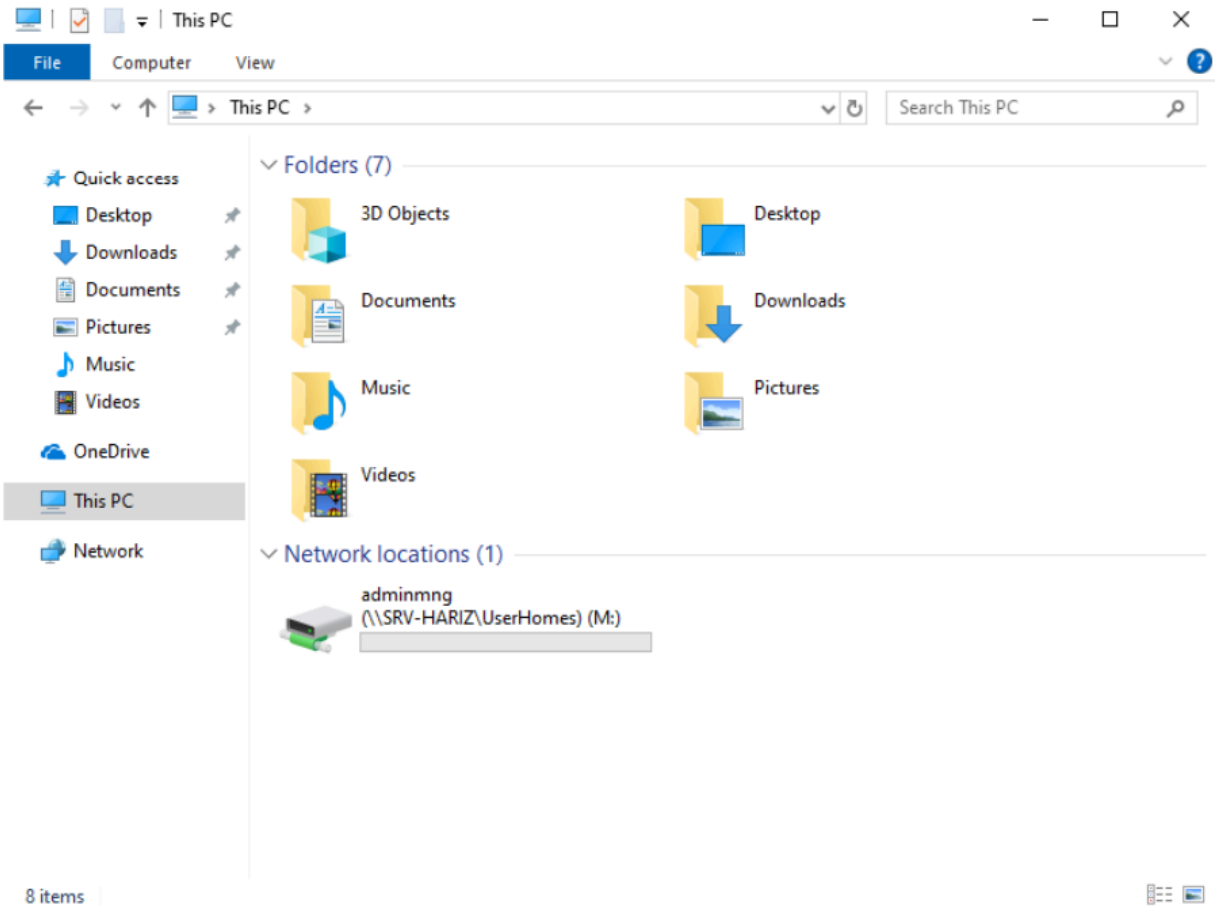
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With GPO, we can also configure that all users cannot see all other drives other than own drive (Home User) :



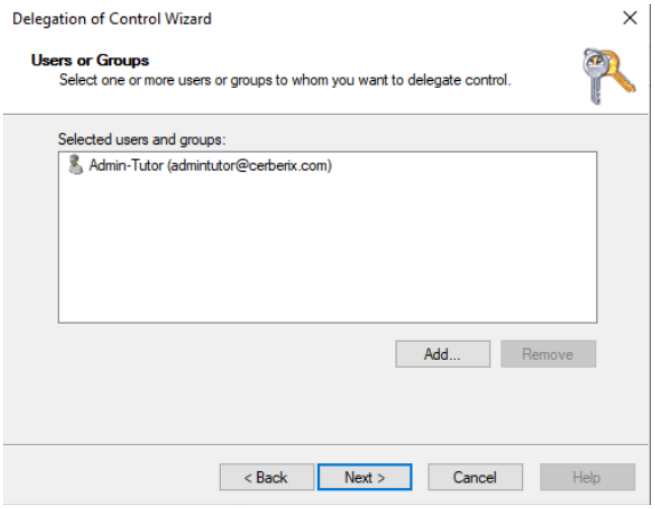
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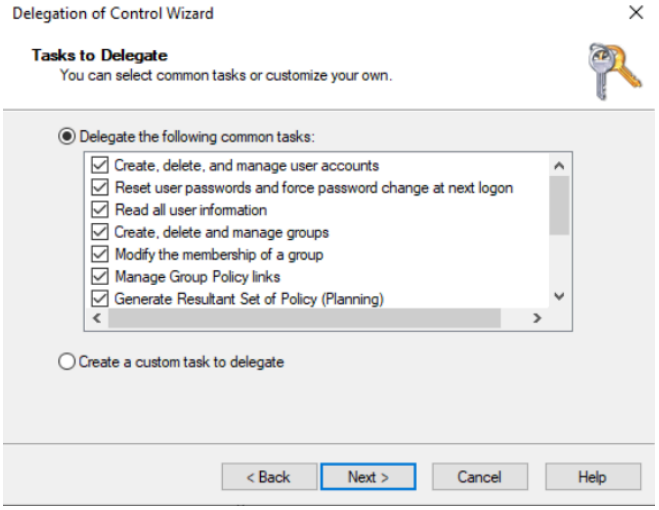
Proof of Concept :



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OU's Administrator



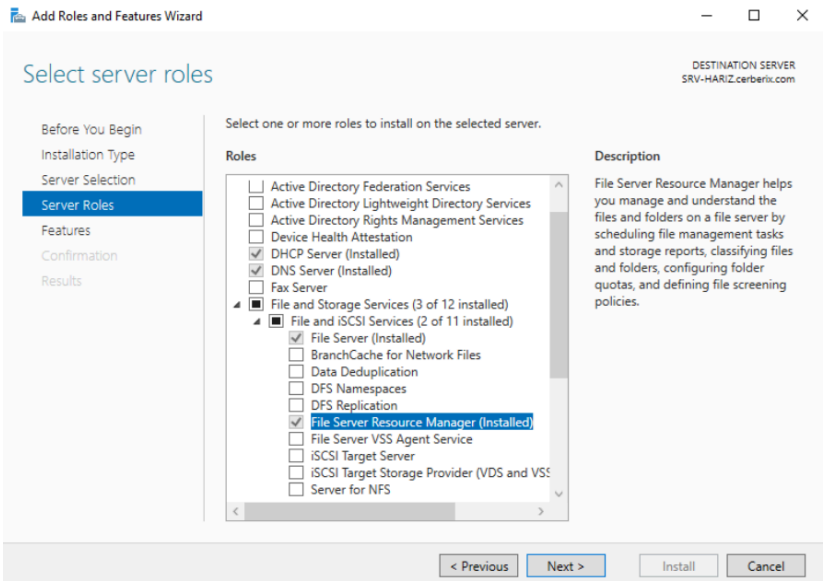


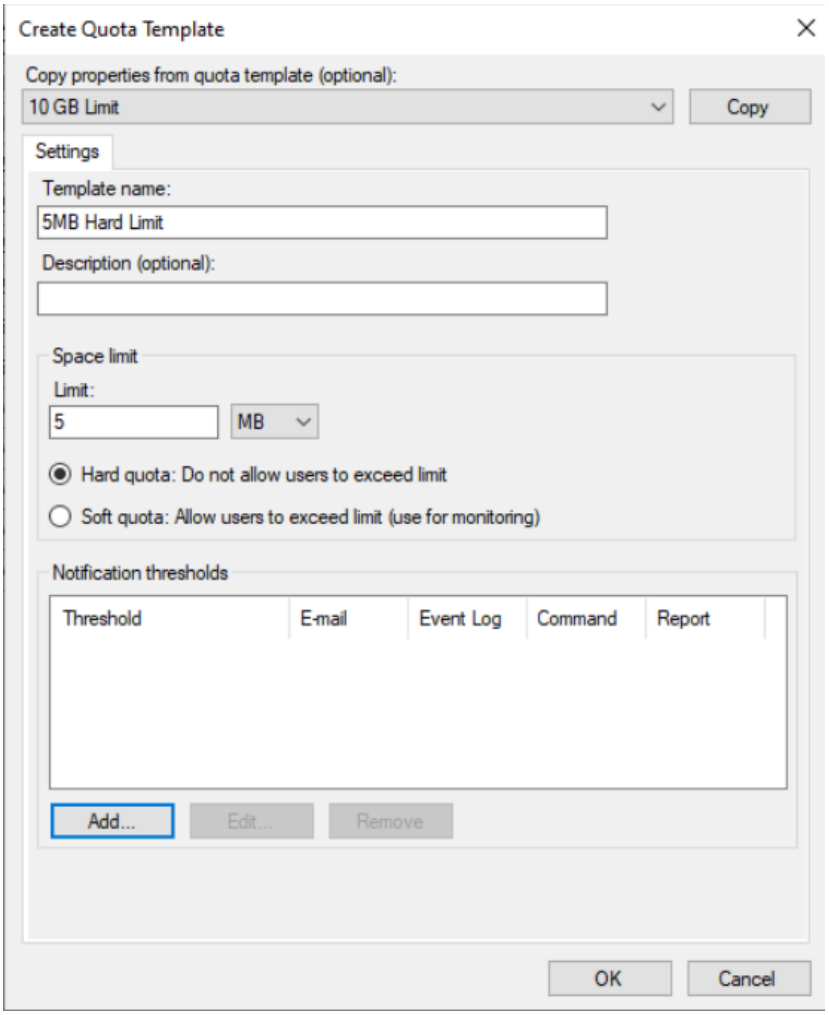


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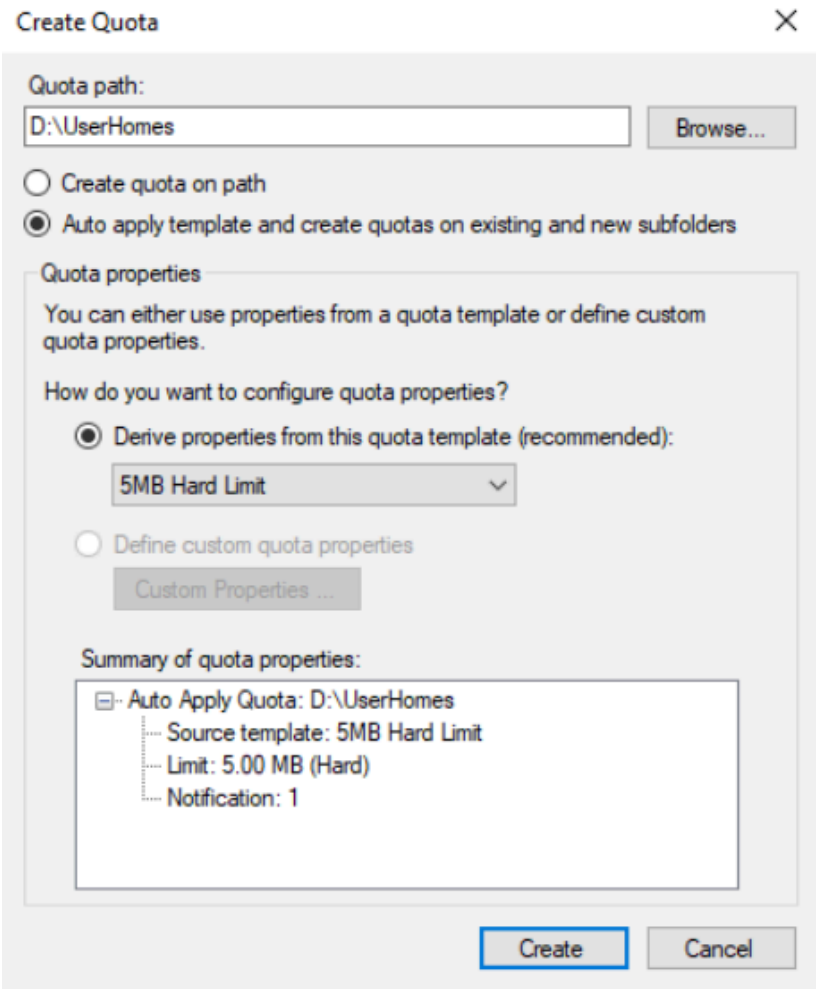
Disk Quota

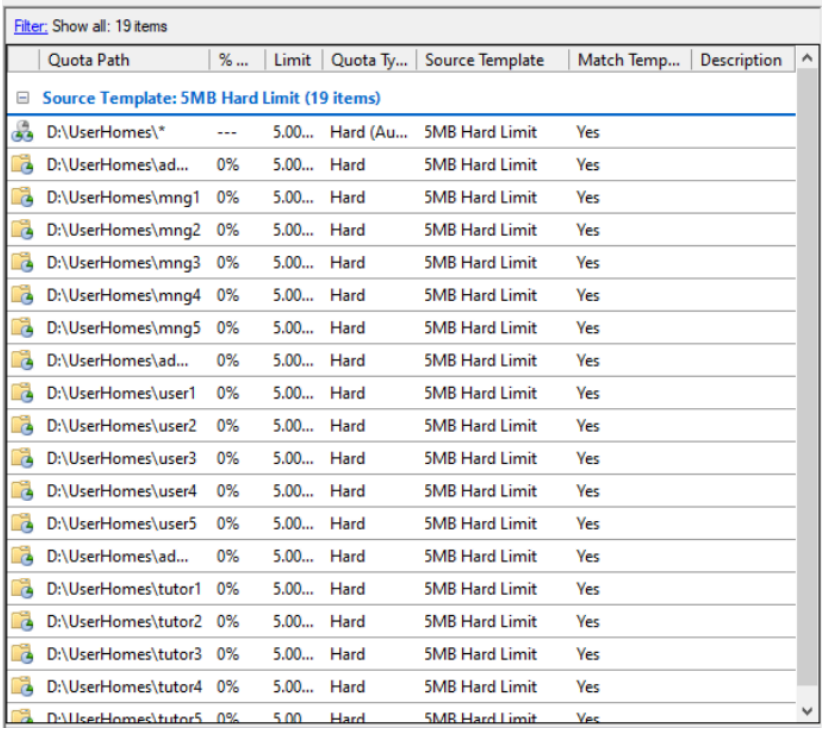
Created a Disk Quota 5MB by installing another service, "File and Storage Services" to manage the quota.





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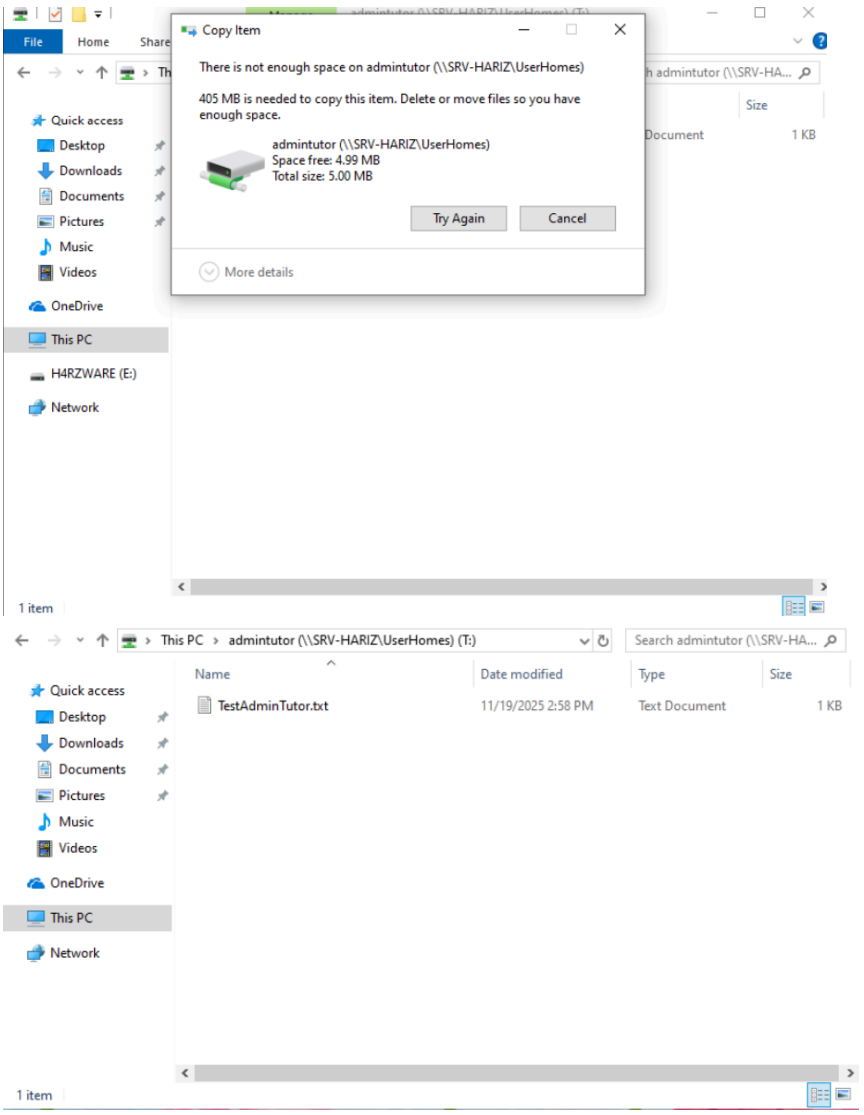
Filter: Show all: 19 items

Quota Path	% ...	Limit	Quota Ty...	Source Template	Match Temp...	Description
Source Template: 5MB Hard Limit (19 items)						
D:\UserHomes*	---	5.00...	Hard (Au...	5MB Hard Limit	Yes	
D:\UserHomes\ad...	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\mng1	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\mng2	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\mng3	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\mng4	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\mng5	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\ad...	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\user1	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\user2	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\user3	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\user4	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\user5	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\ad...	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\tutor1	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\tutor2	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\tutor3	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\tutor4	0%	5.00...	Hard	5MB Hard Limit	Yes	
D:\UserHomes\tutor5	0%	5.00...	Hard	5MB Hard Limit	Yes	

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Proof of concept :

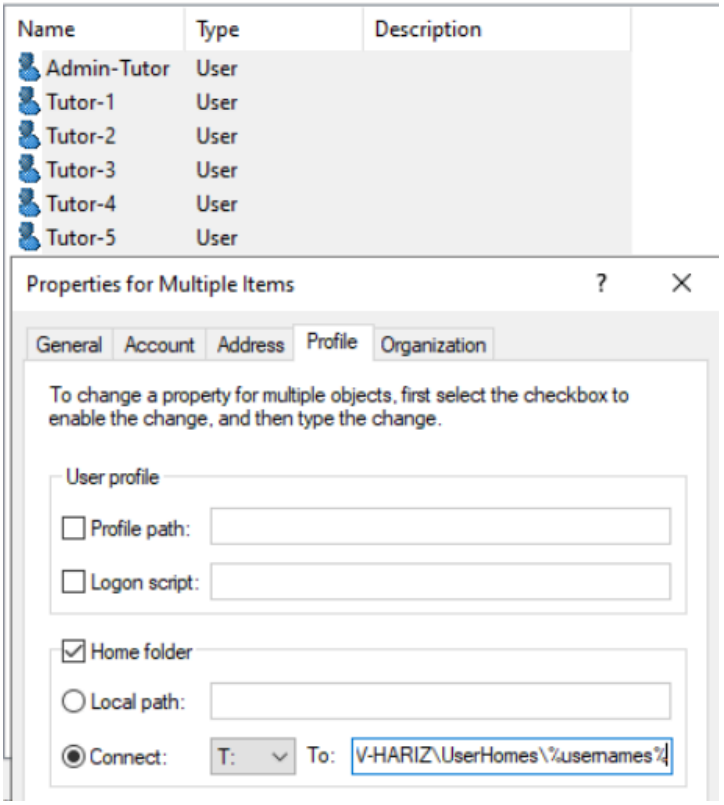
Tested with copying and pasting large files from my pendrive (E:) into the UserHome of Admin-Tutor.

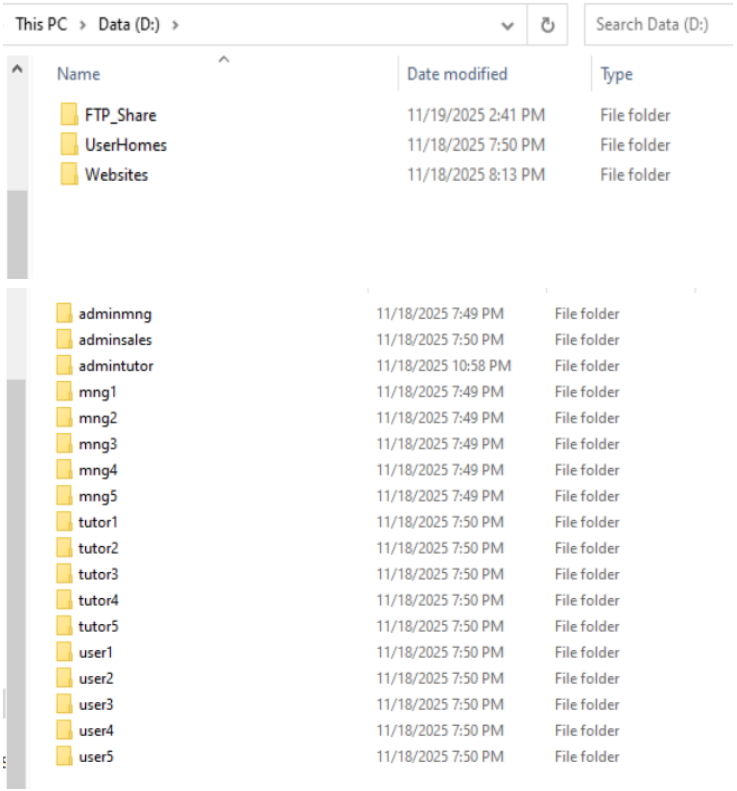


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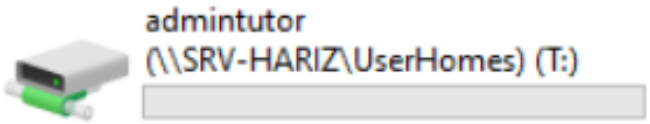
Home Directory

Home directory is assigned by batch.



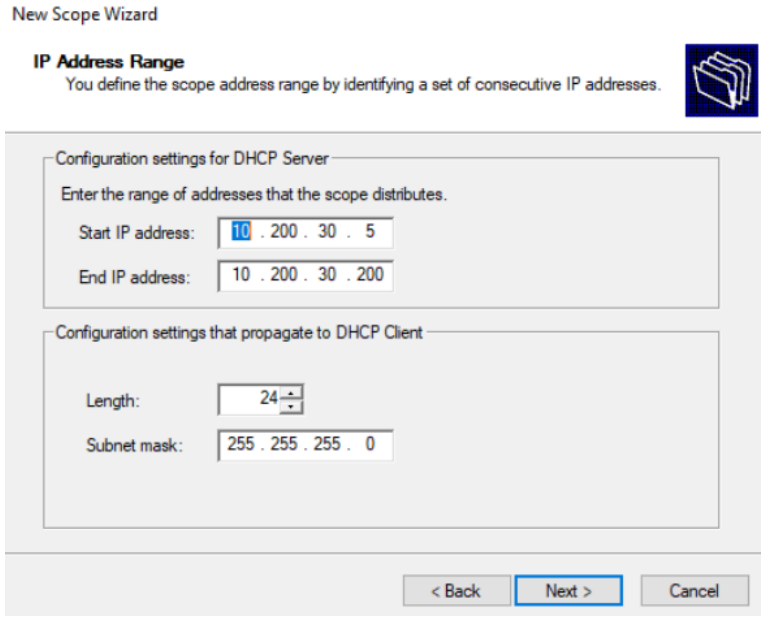


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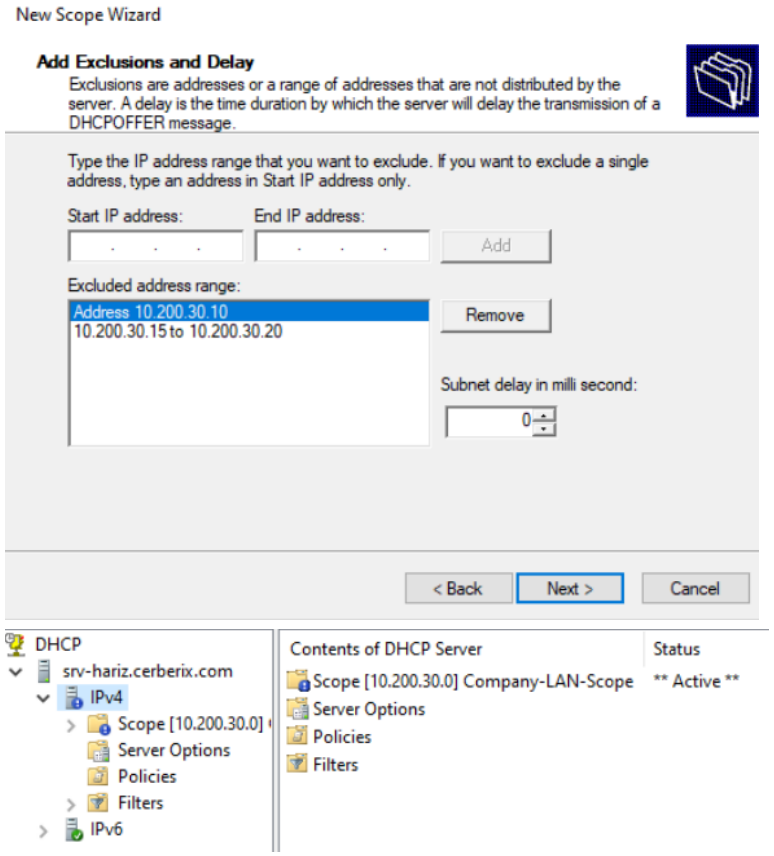


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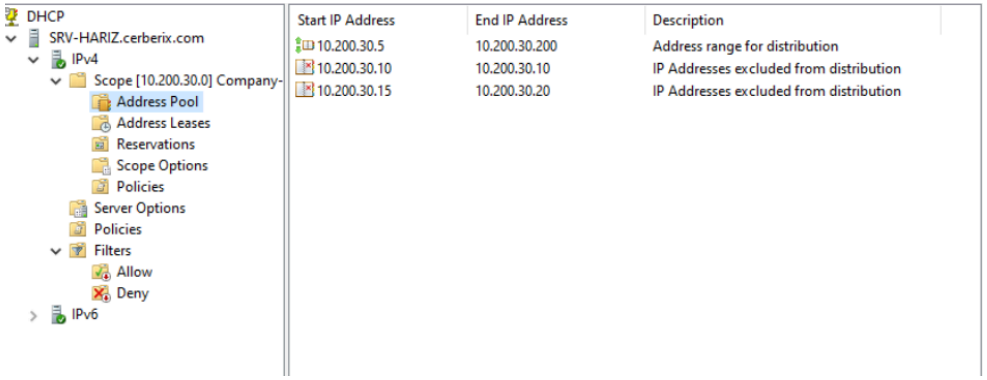
DHCP Configuration

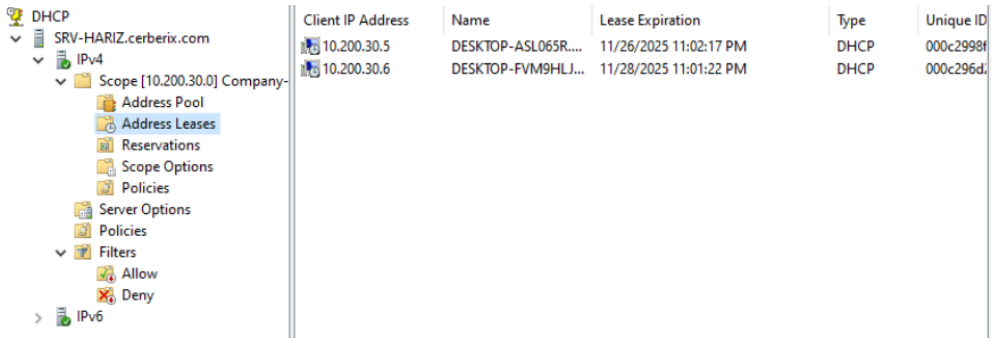


Exclusions | 10.200.30.10 are reserved for the Server and 10.200.30.15 to 10.200.30.20 are spare IP Addresses.



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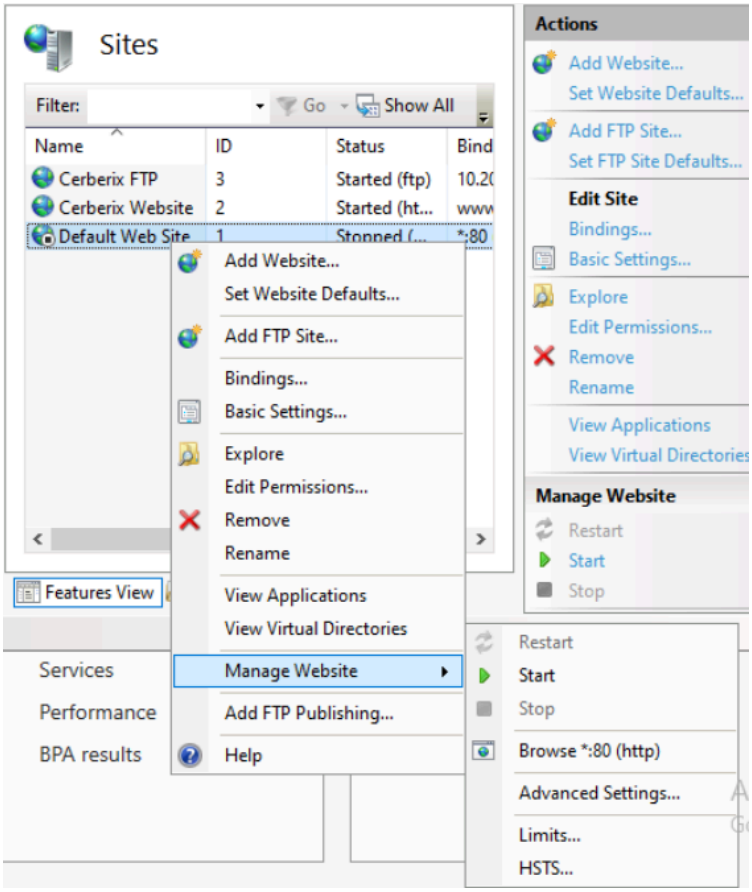




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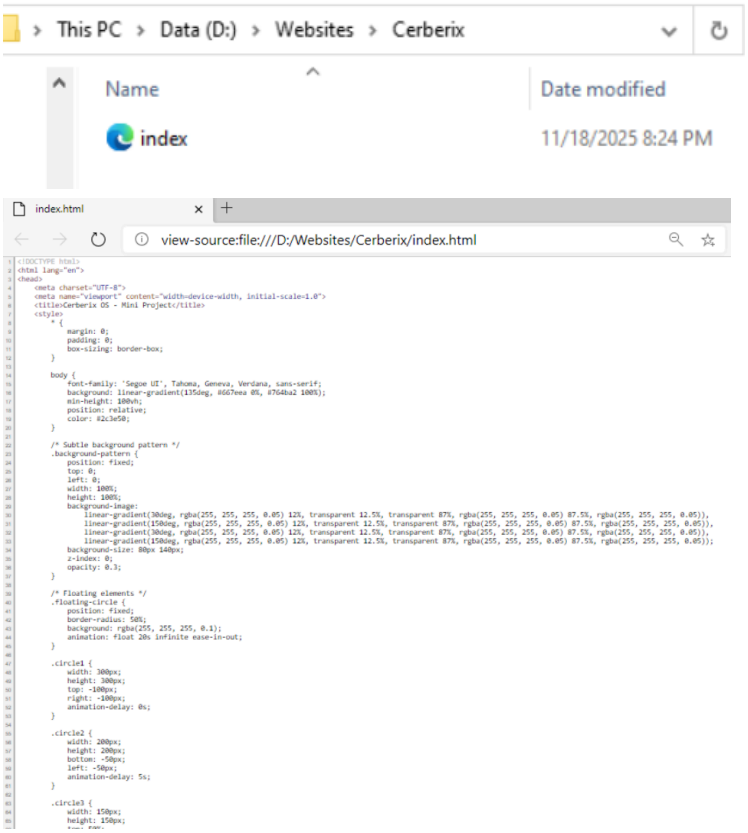
Web IIS

Firstly, we need to "Stop" the Default Website.



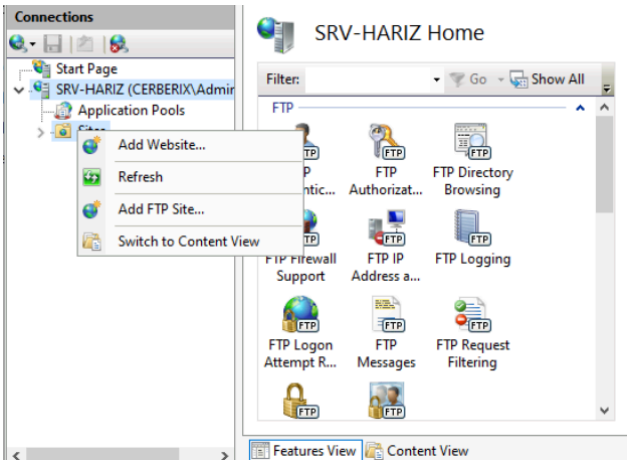
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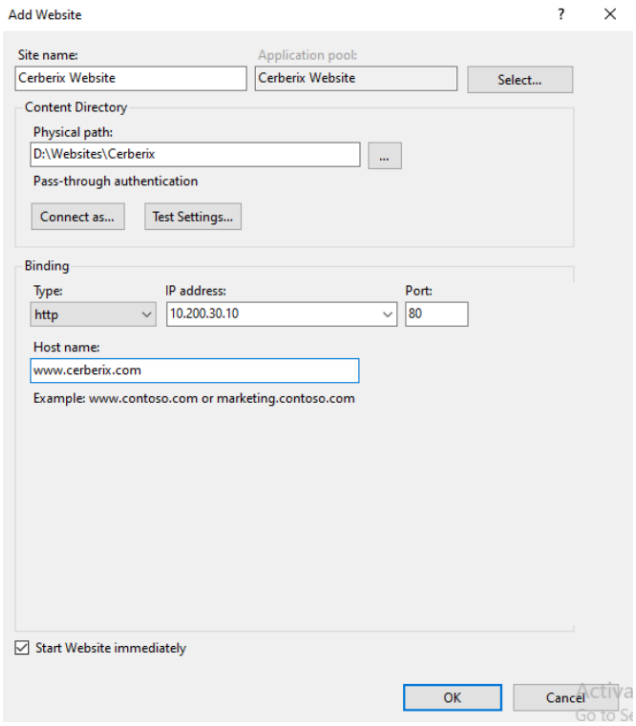
Create a working directory for the web service in the (D:) Drive. For example we did ***/Websites/Cerberix/index.html.*** Below is also the content of index.html

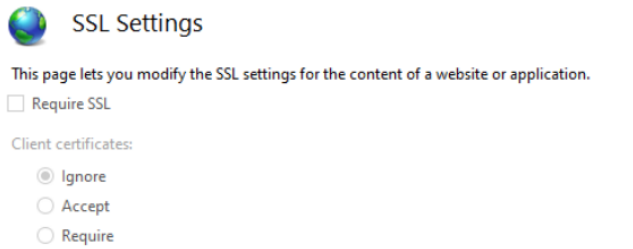


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Then, proceed with setting up our new site in IIS Manager



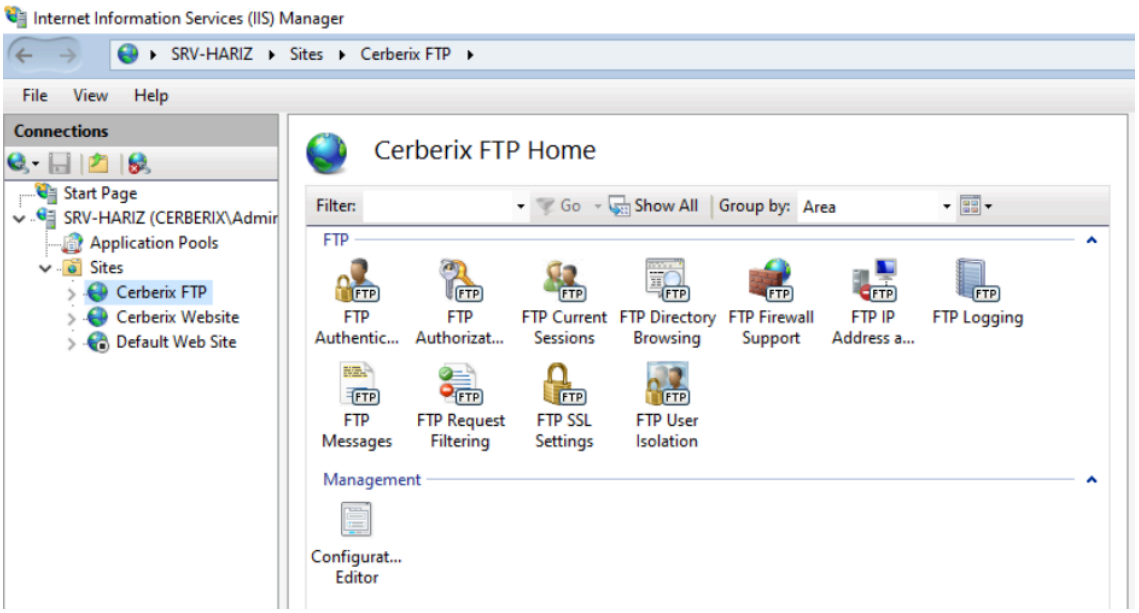


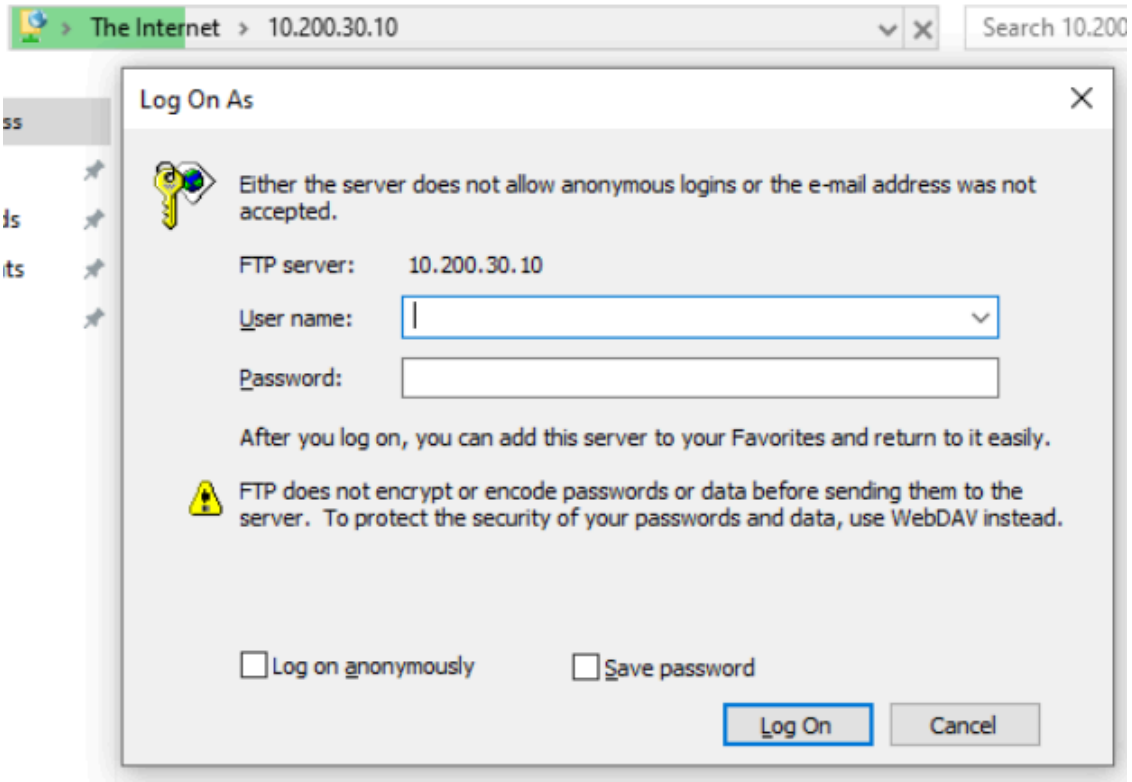


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File Transfer Protocol

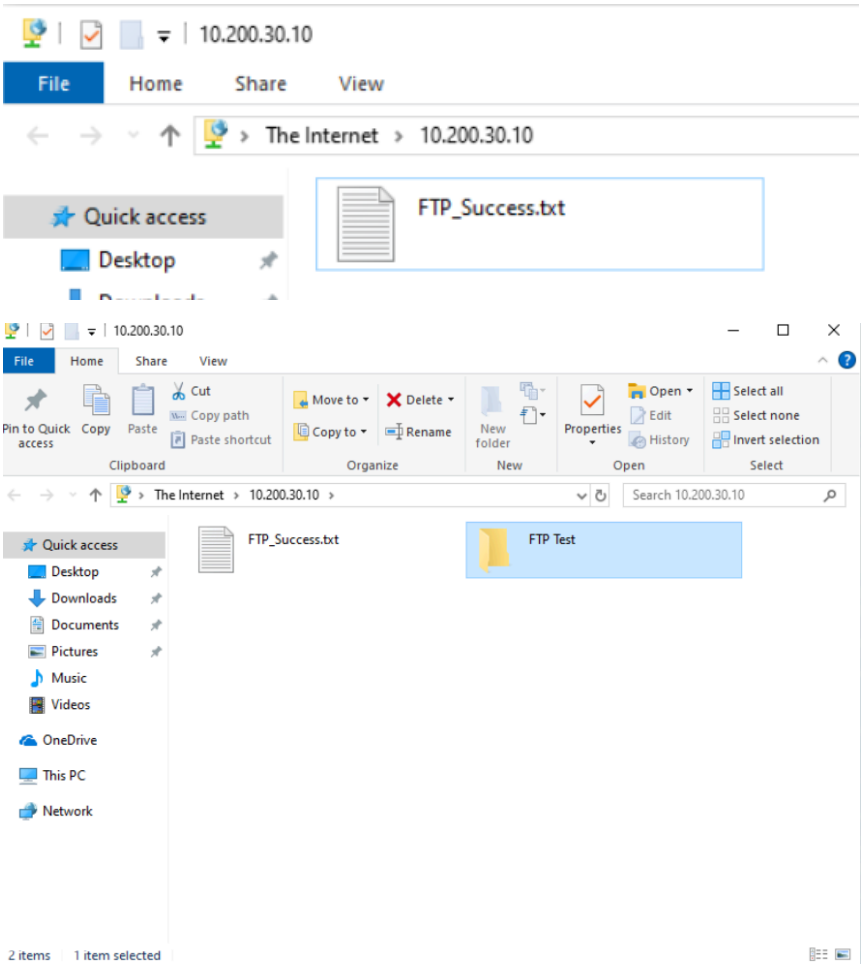
Firstly, install the FTP Server and FTP Service from the IIS. Proceed with creating the FTP Sites from IIS Manager.



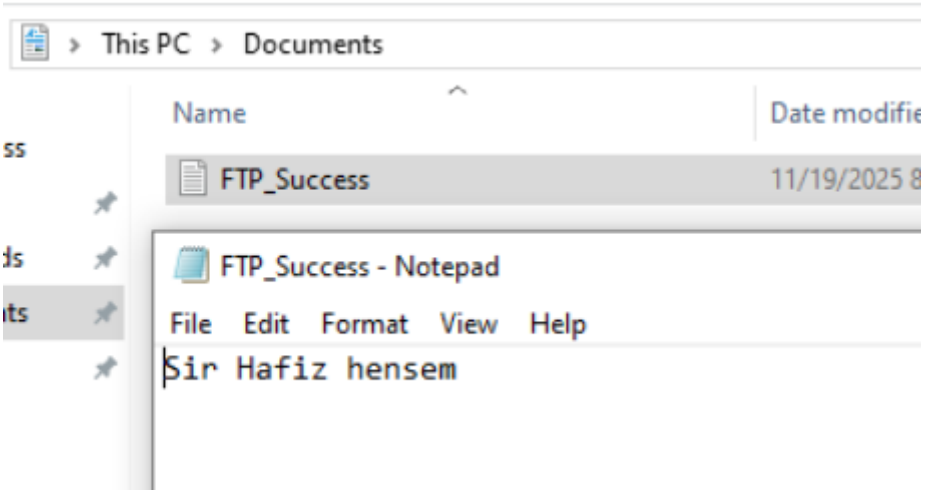


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FTP can be logged on using Authenticated Users creds (Admin-Tutor/Admin-Sales) whichever users are configured in the OU. In this case we are using the creds "admintutor:dcbs_12".



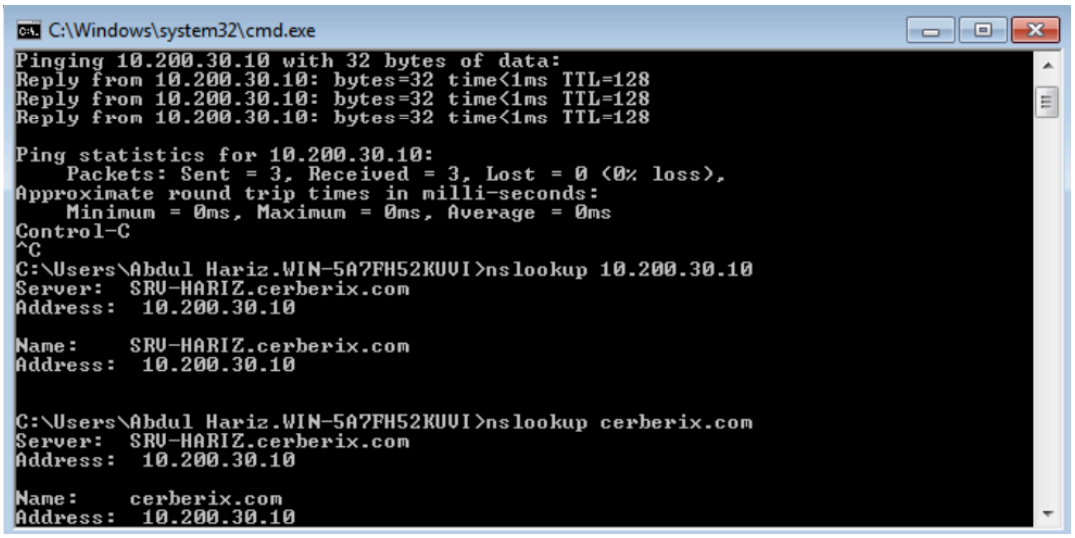
We can only view the file by copying the text file into our windows explorer. Since FTP is on the network, Notepad can't open the text file. Instead it will be opened by Microsoft Edge (Browser). But that also somehow didn't work (the text didn't pop out).



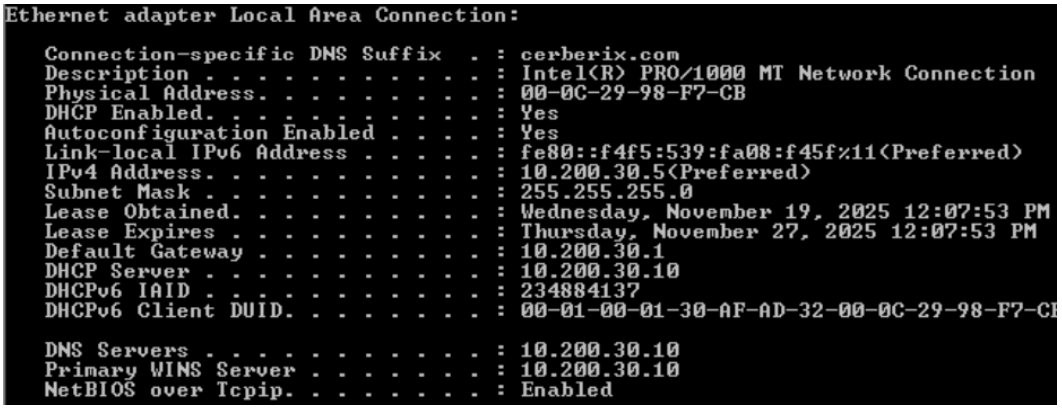
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Client Testing

Pinging and Resolving IP to Domain and Vice Versa (DNS Working) :

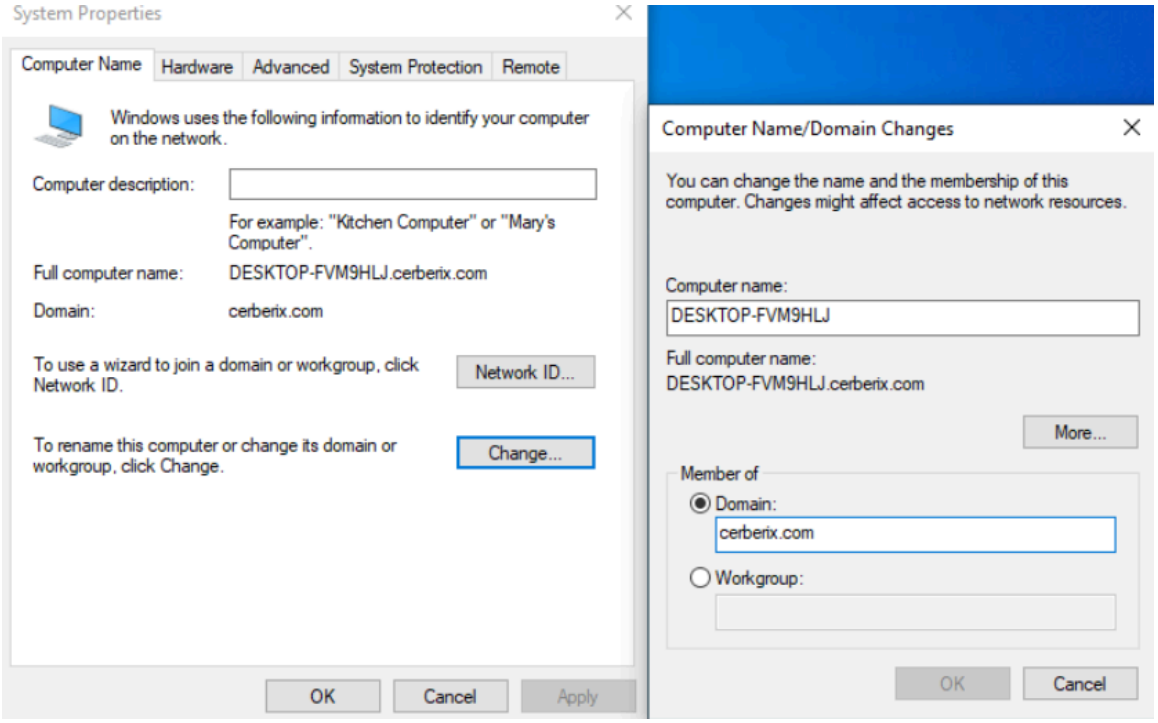


IP Address leased from DHCP Server (DHCP Working) :



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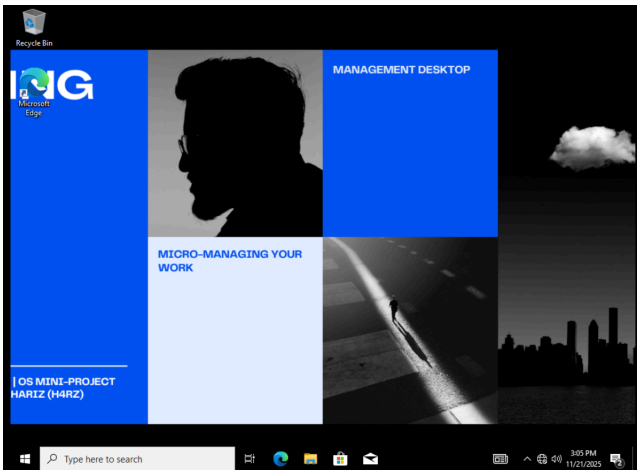
Successfully joined the domain "cerberix.com"



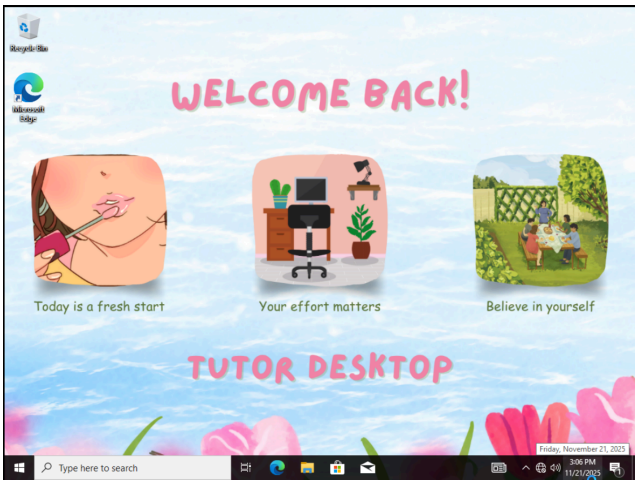
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Personalized Wallpaper :

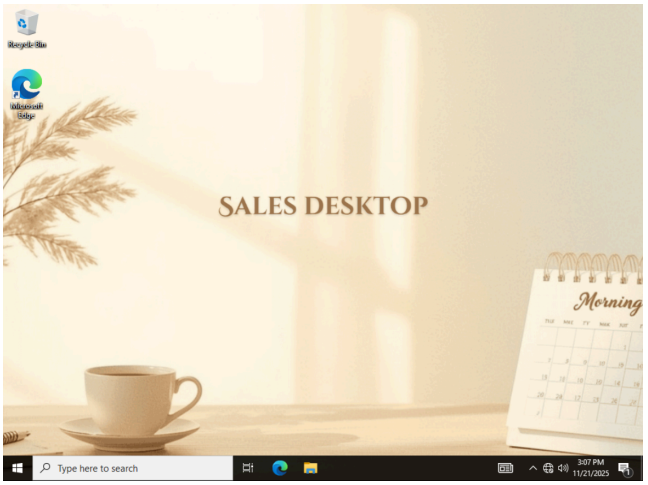
Management Desktop



Tutor Desktop

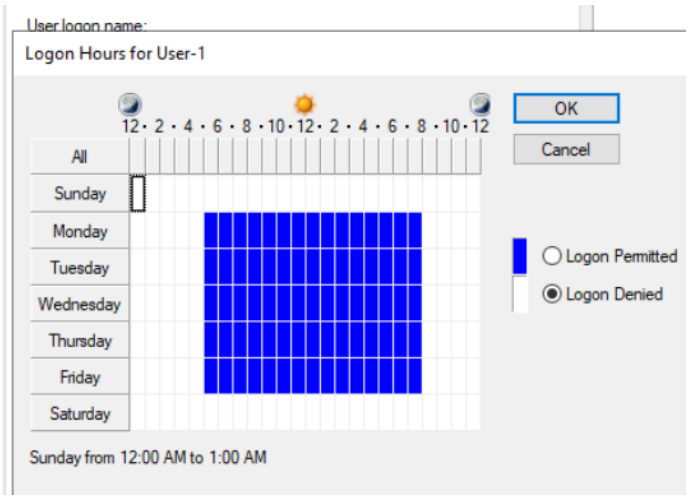


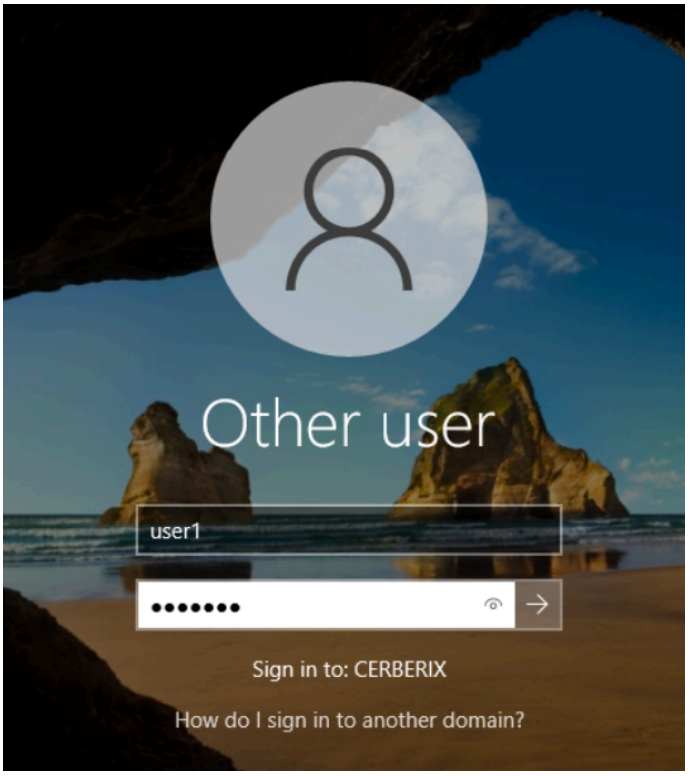
Sales Desktop



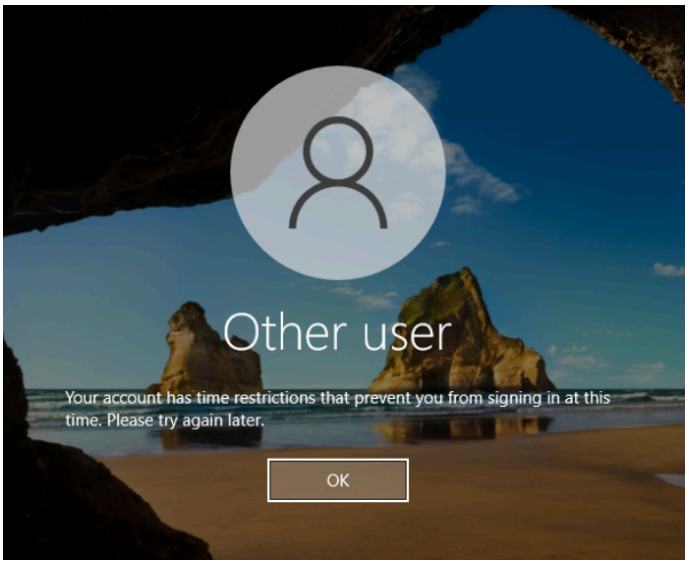
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Logon hours restriction :



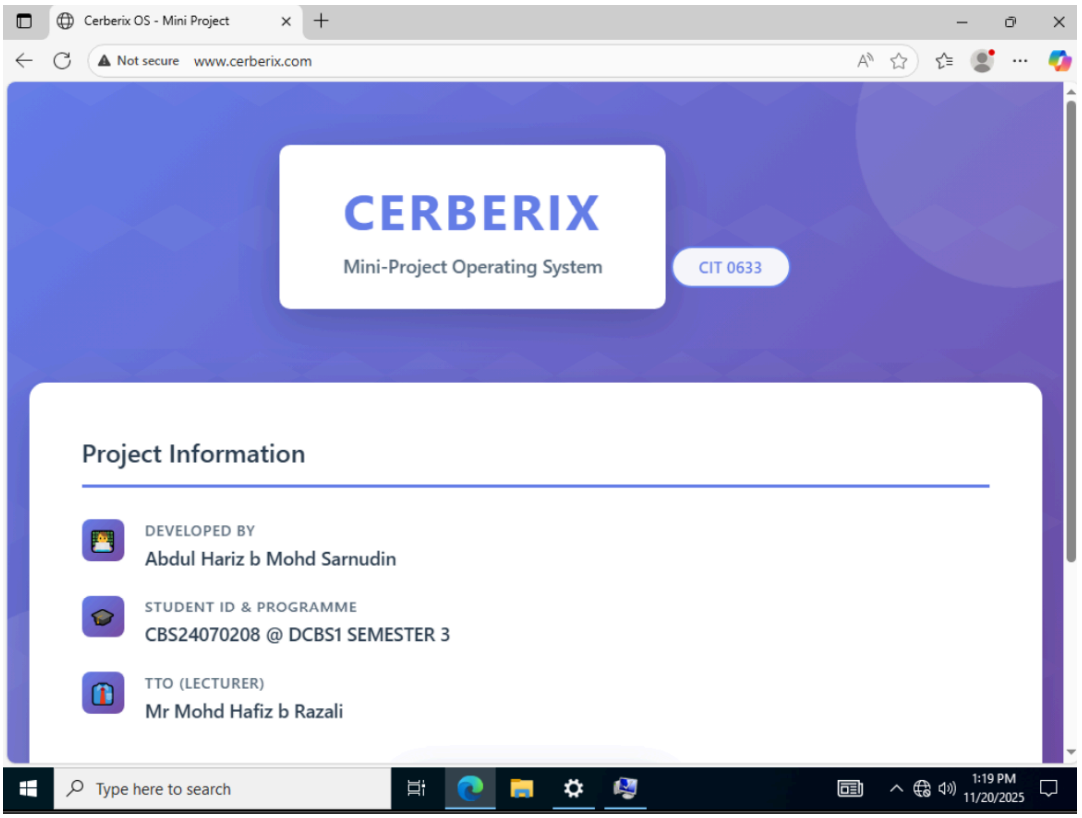


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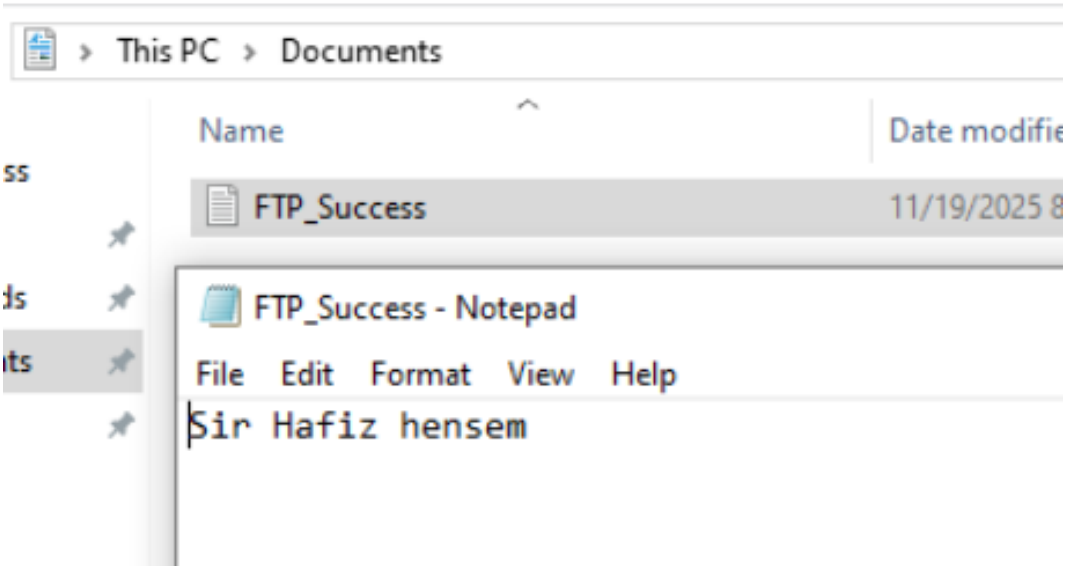


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Website can be access with the Domain we setup, www.cerberix.com and the webpages are accessible (Web IIS Working) :



FTP can be accessed with the authenticated user's credentials (FTP IIS Working) :



END

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